RATIONAL POINTS ON KUMMER VARIETIES AND SWINNERTON-DYER'S METHOD

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Abstract. These lectures are designed to serve as an introduction to Swinnerton-Dyer's descent fibration method. This is a versatile method for studying the Hasse principle and Brauer—Manin obstruction for certain classes of varieties, and works well in situations where the existence of rational points can be related to the existence of rational points on families of torsors under abelian varieties. Following Swinnerton-Dyer's original work, the method has been developed by Colliot-Thélène, Skorobogatov, Harpaz, Wittenberg and others, and examples of varieties it has been successfully applied to include K3 surfaces, cubic surfaces and quartic del Pezzo surfaces. We will primarily illustrate the method in the special case of K3 Kummer surfaces. Time permitting, I will also touch on degree 4 del Pezzo surfaces, which will be discussed in more detail in later lectures of Lyczak. A sizeable portion of the course will be dedicated to the study of 2- and 4-Selmer groups in quadratic twist families, which forms a large part of the technical input to some of the results referenced above.

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1. Introduction

The aim of these notes is to give an introduction to Swinnerton-Dyer's method in the worked example of rational points on Kummer varieties, with a particular focus on (generalized) Kummer surfaces associated to Jacobians of genus 2 curves. A secondary aim is to discuss the behaviour of 2- and 4-Selmer groups in quadratic twist families.

1.1. **Notation/background.** Throughout, k will denote a number field and G_k its absolute Galois group. For a smooth projective variety X over k we have the algebraic Brauer group

$$\mathrm{Br}_1(X) = \ker\big(\mathrm{Br}(X) \to \mathrm{Br}(X_{\bar{k}})\big).$$

We also write $\mathrm{Br}_0(X)$ for the image of $\mathrm{Br}(k)$ in $\mathrm{Br}(X)$. Since $H^3(k, \overline{k}^\times) = 0$ we have an isomorphism $\mathrm{Br}_1(X)/\mathrm{Br}_0(X) \cong H^1\big(k, \mathrm{Pic}(X_{\bar{k}})\big)$; see [CTS21, Proposition 4.3.2] for details.

2. The Cassels–Tate pairing

Let A/k be an abelian variety. The Galois cohomology group $H^1(k, A)$ classifies isomorphism classes of A-torsors X/k. The Shafarevich–Tate group of A is defined as

$$Ш(A) = \ker\left(H^1(k, A) \to \prod_{v \text{ place of } k} H^1(k_v, A)\right).$$

It is the subgroup of $H^1(k, A)$ consisting of elements represented by everywhere locally soluble torsors, i.e. torsors X under A with $X(\mathbb{A}_k) \neq \emptyset$. A major open conjecture is that $Ш(A)$ is finite.

The Cassels–Tate pairing is a bilinear pairing

$$\langle\ ,\ \rangle_{\mathrm{CT}} : Ш(A) \times Ш(A^\vee) \to \mathbb{Q}/\mathbb{Z},$$

where $A^\vee$ denotes the dual abelian variety. It has several equivalent definitions; see [PS99]. One definition, the *homogeneous space definition*, is as follows.

Definition 2.1 (Cassels–Tate pairing)**.** Let X be an everywhere locally soluble A-torsor, so that X represents a class in $Ш(A)$. We have a canonical isomorphism of G_k-modules $\mathrm{Pic}^0(X_{\bar{k}}) \cong \mathrm{Pic}^0(A_{\bar{k}}) = A^\vee(\bar{k})$. This induces a homomorphism

$$\Phi_X : Ш(A^\vee) \to H^1\big(k, \mathrm{Pic}(X_{\bar{k}})\big) \cong \mathrm{Br}_1(X)/\mathrm{Br}_0(X).$$

We note that since elements of $Ш(A^\vee)$ are everywhere locally trivial, the image of this map consists of elements of $\mathrm{Br}_1(X)$ that everywhere locally lie in the image of $\mathrm{Br}(k_v)$ in $\mathrm{Br}(X_{k_v})$. Now take $\eta \in Ш(A^\vee)$ and let $\alpha \in \mathrm{Br}_1(X)$ be any element representing the class of $\Phi_X(\eta)$. Define

$$\langle [X], \eta \rangle_{\mathrm{CT}} = \sum_{v \text{ place of } k} \mathrm{inv}_v \alpha(x_v), \tag{2.2}$$

where $x_v \in X(k_v)$ is any choice of k_v-point on X.

Remark 2.3. The right hand side of (2.2) is independent of the choice of x_v since α is everywhere locally in the image of $\mathrm{Br}(k_v)$, hence the evaluation map $X(k_v) \to \mathrm{Br}(k_v)$ given by $x_v \mapsto \alpha(x_v)$ is constant. Further, by reciprocity for the Brauer group of k, it is independent of the choice of representative $\alpha \in \mathrm{Br}_1(X)$ for $\Phi_X(\eta) \in \mathrm{Br}_1(X)/\mathrm{Br}_0(X)$.

The key property that the Cassels–Tate pairing has is that its left (resp. right) kernel is the maximal divisible subgroup of $Ш(A)$ (resp. $Ш(A^\vee)$). If $Ш(A)$ is finite as expected then so is $Ш(A^\vee)$ and, in particular, the Cassels–Tate pairing is non-degenerate.

In his 1970 ICM talk, Manin defined what is now known as the Brauer–Manin obstruction, and observed the following.

Theorem 2.4 (Manin)**.** *Suppose that $Ш(A)$ is finite. Then the Brauer–Manin obstruction explains all failures of the Hasse principle for torsors under A.*

Proof. Let X be a torsor under A. We can assume that X is everywhere locally soluble, in which case X defines a class $[X] \in Ш(A)$. Suppose that $X(k) = \emptyset$. Then $[X] \neq 0$. Since $Ш(A)$ is finite, the Cassels–Tate pairing is non-degenerate, hence there is $\eta \in Ш(A^\vee)$ with $\langle [X], \eta \rangle_{\mathrm{CT}} \neq 0$. Then any lift $\alpha \in \mathrm{Br}_1(X)$ of $\Phi_X(\eta)$ gives a Brauer–Manin obstruction to the Hasse principle on X. □

2.1. **The obstruction to being alternating.** Suppose now that A admits a principal polarisation $\lambda : A \to A^\vee$. Via λ we identify A with $A^\vee$ and thus view the Cassels–Tate pairing as a $\mathbb{Q}/\mathbb{Z}$-valued bilinear pairing on $Ш(A)$. As shown by Flach, this pairing is antisymmetric. That is, for all $x, y \in Ш(A)$, we have $\langle x, y \rangle_{\mathrm{CT}} = -\langle y, x \rangle_{\mathrm{CT}}$. In particular, one has $2\langle x, x \rangle_{\mathrm{CT}} = 0$ for all $x \in Ш(A)$. One can ask whether, moreover, the Cassels–Tate pairing is alternating (i.e. $\langle x, x \rangle_{\mathrm{CT}} = 0$ for all $x \in Ш(A)$). This question was answered by Poonen and Stoll in [PS99]. We recall their result.

We have a short exact sequence of G_k-modules

$$0 \to A^\vee(\bar{k}) \to \mathrm{Pic}(A_{\bar{k}}) \to NS(A_{\bar{k}}) \to 0.$$

The associated long exact sequence for Galois cohomology induces a connecting map

$$\delta : NS(A_{\bar{k}})^{G_k} \to H^1(k, A^\vee).$$

The principal polarisation λ gives a class in $NS(A_{\bar{k}})^{G_k}$, so we obtain a class $\delta(\lambda) \in H^1(k, A^\vee)$. Denote by c the image of this class under the isomorphism $H^1(k, A^\vee) \cong H^1(k, A)$ induced by λ.

Theorem 2.5 (Poonen–Stoll, [PS99] Theorem 5)**.** *The element c lies in $Ш(A)[2]$. Moreover, for all $x \in Ш(A)$, we have $\langle x, x\rangle_{\mathrm{CT}} = \langle x, c\rangle_{\mathrm{CT}}$. In particular, $\langle\ ,\ \rangle_{\mathrm{CT}}$ is alternating if and only if c lies in the maximal divisible subgroup of $Ш(A)$.*

Corollary 2.6 ([PS99] Theorem 8 and Corollary 9)**.** *Suppose that $Ш(A)$ is finite. If $\langle c, c\rangle_{\mathrm{CT}} = 0$ then $Ш(A) \cong T \times T$ for some finite abelian group T. In particular, $\dim_{\mathbb{F}_2} Ш(A)[2]$ is even and $Ш(A)$ has square order. Conversely, if $\langle c, c\rangle_{\mathrm{CT}} = 1/2$, then $Ш(A) \cong \mathbb{Z}/2\mathbb{Z} \times T \times T$ for some finite abelian group T, $\dim_{\mathbb{F}_2} Ш(A)[2]$ is odd and the order of $Ш(A)$ is twice a square.*

Proof. In general, suppose we have a finite abelian group M, equipped with a non-degenerate antisymmetric bilinear pairing $P : M \times M \to \mathbb{Q}/\mathbb{Z}$. The antisymmetry means that the map $x \mapsto P(x, x)$ is a homomorphism $M \to \frac{1}{2}\mathbb{Z}/\mathbb{Z}$, hence has the form $P(-, c)$ for a unique $c \in M[2]$. The map $\phi : M \to M$ given by

$$m \mapsto \begin{cases} m & P(m, c) = 0, \\ m + c & P(m, c) = 1/2, \end{cases}$$

is a homomorphism. The bilinear pairing $\widetilde{P} : M \times M \to \mathbb{Q}/\mathbb{Z}$ given by $\widetilde{P}(x, y) = P(x, \phi(y))$ is then alternating. Its kernel is $\ker(\phi)$, which is 0 if $P(c, c) = 0$, and is $\langle c\rangle \cong \mathbb{Z}/2\mathbb{Z}$ if $P(c, c) = 1/2$. In this latter case, we have $M \cong \langle c\rangle \oplus \langle c\rangle^{\perp}$, where the orthogonal complement is taken with respect to P and, moreover, $\widetilde{P}$ is a non-degenerate alternating pairing on $\langle c\rangle^{\perp}$. Recall that any finite abelian group admitting a non-degenerate alternating pairing is isomorphic to $T \times T$ for some finite abelian group T. We deduce that there is a finite abelian group T such that:

$$M \cong \begin{cases} T \times T & P(c, c) = 0, \\ \mathbb{Z}/2\mathbb{Z} \times T \times T & P(c, c) = 1/2. \end{cases}$$

Applying this with $M = Ш(A)$ and P the Cassels–Tate pairing gives the result. □

The class $\delta(\lambda) \in H^1(k, A^\vee)$ is the $A^\vee$-torsor associated to the G_k-set of line bundles $\mathcal{L} \in \mathrm{Pic}(A_{\bar{k}})$ inducing the given principal polarisation λ (that is, such that λ is given by $x \mapsto \tau_x^*\mathcal{L} \otimes \mathcal{L}^{-1}$ where $\tau_x : A \to A$ is translation by x). This admits a natural refinement to an $A^\vee[2]$-torsor. Namely, a lift of $\delta(\lambda)$ to $H^1(k, A^\vee[2])$ is given by the class associated to the G_k-set of symmetric line bundles inducing λ. This gives a useful interpretation of the Poonen–Stoll class c as follows; see [PR11, Section 3.4] for details. Via the principal polarisation λ, we view the Weil pairing $e_2 : A[2] \times A^\vee[2] \to \mu_2$ as a pairing $A[2] \times A[2] \to \mu_2$, which by an abuse of notation we denote by e_2 also. It is non-degenerate, bilinear, alternating and G_k-equivariant. In particular, it is antisymmetric hence (since $A[2]$ is 2-torsion) symmetric. A *quadratic refinement* of e_2 is a map $q : A[2] \to \mu_2$ such that

$$q(x + y)q(x)^{-1}q(y)^{-1} = e_2(x, y)$$

for all $x, y \in A[2]$. The G_k-set of quadratic refinements of the Weil pairing on $A[2]$ is a principal homogeneous space for $A[2]$, hence gives rise to a class $c_2 \in H^1(k, A[2])$. This class lifts $c \in H^1(k, A)$ to $H^1(k, A[2])$. Recall that, by definition, the preimage of $Ш(A)[2]$ in $H^1(k, A[2])$ is the 2-Selmer group $\mathrm{Sel}_2(A)$. We deduce the following result from the above discussion:

Proposition 2.7. *The class c_2 lies in $\mathrm{Sel}_2(A)$. Moreover, writing $G = \mathrm{Gal}(k(A[2])/k)$, the class c_2 arises via inflation from $H^1(G, A[2])$.*[1] *In particular, if G_k acts on $A[2]$ through the orthogonal group $O(q)$ of a quadratic refinement q of the Weil pairing (e.g. if $A[2] \subseteq A(k)$) then $c = 0$. In this case, if $Ш(A)$ is finite then $\dim_{\mathbb{F}_2} Ш(A)[2]$ is even.*

[1] Since the Galois action on $A[2]$ determines the Galois action on the set of quadratic refinements of e_2.

2.2. **Quadratic twists.** Let $\chi : G_k \to \mu_2$ be a quadratic character. Denote by A^χ the quadratic twist of A by χ. That is, A^χ is the twist of A given by viewing χ as a 1-cocycle valued in $\{\pm 1\} \subseteq \mathrm{Aut}(A_{\bar{k}})$. Thus, A^χ is equipped with a $\bar{k}$-isomorphism $\phi : A \to A^\chi$ such that $\phi^{-1} \circ \sigma\phi\sigma^{-1}$ is multiplication by $\chi(\sigma)$. In particular, A^χ is isomorphic to A over the quadratic extension cut out by χ.

Note that ϕ gives a Galois equivariant isomorphism $A[2] \cong A^\chi[2]$. Via this map, we will always identify $A^\chi[2]$ with $A[2]$ in what follows. The principal polarisation on A descends via ϕ to a principal polarisation on A^χ (since multiplication by -1 acts trivially on the Néron–Severi group of A) and the identification of $A[2]$ with $A^\chi[2]$ is compatible with the resulting Weil pairings.

Remark 2.8. Proposition 2.7 is particularly useful when considering the behaviour of Selmer/Shafarevich–Tate groups under quadratic twist. Indeed, if A' is a quadratic twist of A, then A' is also principally polarised (more precisely, the given polarisation λ descends to a principal polarisation on A') and there is an isomorphism of Galois modules $A[2] \cong A'[2]$ identifying the respective Weil pairings. This shows, for example, that if G_k acts on $A[2]$ through the orthogonal group $O(q)$ of a quadratic refinement q of the Weil pairing, and the Shafarevich–Tate groups of all quadratic twists of A are finite, then $\dim_{\mathbb{F}_2} Ш(A')[2]$ is even for every quadratic twist A' of A.

3. Remarks on Swinnerton-Dyer's method for Kummer varieties

The following is a situation in which one might hope that the existence of rational points on a given variety can be related to the existence of rational points on torsors under abelian varieties.

3.1. **Generalised Kummer varieties.** Let A/k be a principally polarised abelian variety of dimension at least 2. The Kummer variety X associated to A is a desingularisation of the quotient $X_{\mathrm{sing}} = A/\pm 1$. More specifically, X_{sing} is singular at the image of $A[2]$ and X is the blow up of X_{sing} at these points. When A has dimension 2, X is a $K3$ surface. Note that $X(k) \neq \emptyset$ since the same is true of A. However, there is a generalisation of this construction that yields varieties where the existence of rational points is much more interesting.

Fix a non-zero class $\alpha \in H^1(k, A[2])$ and denote by Y_α the corresponding 2-covering of A. Thus, Y_α is a torsor under A equipped with an involution ι compatible with multiplication by -1 on A. The generalised Kummer variety X_α associated to α is the blow up of the quotient $Y_\alpha / \langle \iota \rangle$ at the image of the fixed points of ι. It is a twist of X. Denote by $\widetilde{Y}_\alpha$ the blow up of Y_α at the fixed points of ι. Then ι extends uniquely to an involution $\widetilde{\iota}$ on $\widetilde{Y}_\alpha$ and $X_\alpha = \widetilde{Y} / \langle \widetilde{\iota} \rangle$.

Now let $\chi : G_k \to \mu_2$ be a quadratic character. We identify μ_2 with $\langle \iota \rangle \leq \mathrm{Aut}_{\bar{k}}(Y_\alpha)$ and thus view χ as a 1-cocycle valued in $\mathrm{Aut}_{\bar{k}}(Y_\alpha)$. Denote by Y_α^χ the corresponding quadratic twist of Y_α by χ. It is the 2-covering for A^χ arising from viewing α as an element of $H^1(k, A^\chi[2])$ via the identification $A[2] \cong A^\chi[2]$ discussed above. Since the construction of X_α involves taking the quotient by $\langle \iota \rangle$, the Kummer variety associated to Y_α^χ is (isomorphic over k to) X_α. Thus, for each quadratic character χ, we have a degree 2 morphism $\widetilde{Y}_\alpha^\chi \to X_\alpha$. Note, in particular, that if $Y_\alpha^\chi(k) \neq \emptyset$ for some χ then $X_\alpha(k) \neq \emptyset$. Conversely, if $X_\alpha(k) \neq \emptyset$, say $x \in X_\alpha(k)$, then the preimage of x under the quotient map $f : \widetilde{Y}_\alpha \to X_\alpha$ has order at most 2 and is stable under G_k. We deduce from this that, after a suitable quadratic twist, the fibre over x is defined over k. That is, there is a quadratic character χ such that $\widetilde{Y}_\alpha^\chi(k) \neq \emptyset$, which is the case if and only if $Y_\alpha^\chi(k) \neq \emptyset$. In this way, the existence of a k-point on X_α is equivalent to the existence of a k-point on Y_α^χ for some χ, reducing the study of rational points on X_α to the study of rational points in a family of torsors under abelian varieties.

3.2. **Basic strategy.** By a conjecture of Skorobogatov, one expects that the Brauer–Manin obstruction explains all failures of the Hasse principle for $K3$ surfaces, hence in particular Kummer surfaces. The Brauer group of generalised Kummer varieties is studied by Skorobogatov–Zarhin [SZ17]. Suppose we are in a situation where (we do not expect) the Brauer group to obstruct the Hasse principle on X_α. Suppose also that $X_\alpha(\mathbb{A}_k) \neq \emptyset$. From the discussion above this implies, in particular, that for each place v of k, there is a quadratic character $\chi_v : G_{k_v} \to \mu_2$ such that $Y_\alpha^{\chi_v}(k_v) \neq \emptyset$. The basic

strategy (which often turns out to be too optimistic) for establishing the existence of a k-point on X_α conditional on finiteness of Shafarevich–Tate groups, goes as follows:

- find a global quadratic character $\chi : G_k \to \mu_2$ such that $Y_\alpha^\chi(\mathbb{A}_k) \neq \emptyset$. Equivalently, such that $\alpha \in \mathrm{Sel}_2(A^\chi)$.
- construct a sequence $\chi = \chi_1, \chi_2, ..., \chi_n$ of quadratic characters such that passing from χ_i to χ_{i+1} induces a controlled change on the 2-Selmer group, say
$$\mathrm{Sel}_2(A^\chi) \supsetneq \mathrm{Sel}_2(A^{\chi_2}) \supsetneq \mathrm{Sel}_2(A^{\chi_3}) \supsetneq \cdots \supsetneq \mathrm{Sel}_2(A^{\chi_n}) = \langle \alpha \rangle .$$
The idea here is to alter χ in such a way that α remains in the 2-Selmer group at each step, but such that the 2-Selmer group of $A^{\chi_{i+1}}$ is strictly smaller than that of A^{χ_i}. If one can always do this, one eventually arrives at a situation where $\mathrm{Sel}_2(A^{\chi_n})$ is 1-dimensional, generated by α (this can only be possible when $A(k)[2] = 0$, but one can consider variants where $\mathrm{Sel}_2(A^{\chi_n})$ is generated by α and the image of the k-rational 2-torsion points, etc.).
- We have a surjective map $\mathrm{Sel}_2(A^{\chi_n}) \to Ш(A^{\chi_n})[2]$. If we are in a situation where finiteness of $Ш(A^{\chi_n})$ implies that $Ш(A^{\chi_n})[2]$ is even dimensional (e.g. if G_k acts on $A[2]$ through the orthogonal group of some quadratic refinement of the Weil pairing) then, conditional on finiteness of Shafarevich–Tate groups, we must have $Ш(A^{\chi_n})[2] = 0$. Thus the class of α in $Ш(A^{\chi_n})$ is trivial, hence $Y^\alpha(k) \neq \emptyset$. Thus also $X_\alpha(k) \neq \emptyset$.

We will see later some situations where the strategy works in exactly this way. However, there are many others where this strategy runs into difficutly and additional tricks must be found to save it. A key recent innovation, due to Harpaz, is to incorporate a second descent step (more about this later).

4. Selmer groups

We begin with the abstract famework of Selmer structures, before specialising to Selmer groups of abelian varieties.

4.1. **Selmer structures.** Let M be a finite G_k-module with exponent n. Define the *dual* of M, $M^* = \mathrm{Hom}(M, \mu_n)$. Define, for v a non-archimedean place of K, the *unramified subspace*
$$H^1_{\mathrm{ur}}(k_v, M) := \ker\left(H^1(k_v, M) \xrightarrow{\mathrm{res}} H^1(k_v^{\mathrm{ur}}, M)\right),$$
where k_v^{ur} denotes the maximal unramified extension of k_v. The inflation-restriction exact sequence means that each class in $H^1_{\mathrm{ur}}(k_v, M)$ can be represented by a 1-cocycle that factors through $\mathrm{Gal}(k_v^{\mathrm{ur}}/k_v)$.

For each place v of k, define the *local Tate pairing*
$$\langle\ ,\ \rangle_v : H^1(k_v, M) \times H^1(k_v, M^*) \longrightarrow H^2(k_v, \mu_n) = \mathrm{Br}(K_v)[2] \hookrightarrow \mathbb{Q}/\mathbb{Z}$$
given by the composition of cup-product and the local invariant map.

Theorem 4.1 (Tate local duality)**.** *For each place v of k, the local Tate pairing is non-degenerate. Moreover, for each nonarchimedean place $v \nmid n$ such that the inertia group acts trivially on M, $H^1_{\mathrm{ur}}(k_v, M)$ and $H^1_{\mathrm{ur}}(k_v, M^*)$ are orthogonal complements.*

Proof. See [NSW08, Corollary 7.2.6] for non-archimedean v and op. cit. Theorem 7.2.17 for archimedean v. The claim about unramified subspaces is op. cit. Theorem 7.2.15. □

Definition 4.2 (Selmer structure)**.** A *Selmer structure* $\mathcal{L} = \{\mathcal{L}_v\}_v$ for M is a collection of subspaces $\mathcal{L}_v \subseteq H^1(k_v, M)$ for each place v of k, such that $\mathcal{L}_v = H^1_{\mathrm{ur}}(k_v, M)$ for all but finitely many v. The associated *Selmer group* $\mathrm{Sel}_{\mathcal{L}}(M)$ is defined as
$$\mathrm{Sel}_{\mathcal{L}}(M) = \{x \in H^1(k, M) \ : \ \mathrm{res}_v(x) \in \mathcal{L}_v \text{ for all places } v \text{ of } k\}.$$
The assumption that $\mathcal{L}_v = H^1_{\mathrm{ur}}(k_v, M)$ for all but finitely many places ensures that $\mathrm{Sel}_{\mathcal{L}}(M)$ is finite. Define the *dual Selmer structure* $\mathcal{L}^* = \{\mathcal{L}_v^*\}$ for M^* by taking $\mathcal{L}_v^*$ to be the orthogonal complement of $\mathcal{L}_v$ for each place v. We refer to $\mathrm{Sel}_{\mathcal{L}^*}(M)$ as the *dual Selmer group*.

4.2. **Selmer groups of abelian varieties.** Let A be a principally polarised abelian variety over k. We now specalise to the case that $M = A[n]$ for some integer $n \geq 2$. The assumption that A is principally polarised means that $A[n]$ is self-dual via the Weil pairing. For each place v, taking Galois cohomology of the short exact sequence

$$0 \to A[n] \to A \xrightarrow{n} A \to 0 \tag{4.3}$$

induces a connecting map $\delta : A(k_v)/nA(k_v) \hookrightarrow H^1(k_v, A[n])$.

Lemma 4.4. *Let v be a place of k.*

(i) *$\delta(A(k_v))$ is its own orthogonal complement with respect to the local Tate pairing,*
(ii) *Suppose that $v \nmid n$ is a nonarchimedean place of k. Then*

$$\#\delta(A(k_v)) = \#A(k_v)[n].$$

Moreover, if A has good reduction at v, then $\delta(A(k_v)) = H^1_{\mathrm{ur}}(k_v, A[n])$.

As a result of Lemma 4.4, setting $\mathcal{L}_v = \delta(A(k_v))$ gives a self-dual Selmer structure for $A[n]$. The resulting Selmer group is the n-Selmer group of A, $\mathrm{Sel}_n(A) \subseteq H^1(k, A[n])$. Taking Galois cohomology of the sequence (4.3) over both k and k_v, we deduce the existence of a the well-known short exact sequence

$$0 \to A(k)/nA(k) \to \mathrm{Sel}_n(A) \to Ш(A)[n] \to 0.$$

4.3. 2-**Selmer groups in quadratic twist families.** Let $\chi : G_k \to \mu_2$ be a quadratic character. As above, we identify $A[2]$ with $A^\chi[2]$ as G_k-modules. In particular, we view $\mathrm{Sel}_2(A^\chi)$ as a subgroup of $H^1(k, A[2])$. We can thus hope to compare $\mathrm{Sel}_2(A)$ and $\mathrm{Sel}_2(A^\chi)$ by comparing the relevant local conditions subgroups. Note that since the isomorphism $A[2] \cong A^\chi[2]$ is compatible with the Weil pairing, it identifies the local Tate pairings also.

Let v be a place of k. We will denote by δ both the map $A(k_v)/2A(k_v) \to H^1(k_v, A[2])$ considered previously, and the map $A^\chi(k_v)/2A^\chi(k_v) \to H^1(k_v, A^\chi[2]) \cong H^1(k_v, A[2])$ given by composing the connecting map for A^χ with the isomorphism $A^\chi[2] \cong A[2]$ induced by ϕ^{-1}. We thus have two self-dual Selmer structures $\{\delta(A(k_v))\}_v$ and $\{\delta(A^\chi(k_v))\}_v$ on $A[2]$. Note that restricting these connecting maps to $A[2] \cong A^\chi[2]$ gives two different connecting maps $\delta : A(k_v)[2] \to H^1(k, A[2])$. The first is the connecting map associated to the short exact sequence $0 \to A[2] \to A[4] \xrightarrow{2} A[2] \to 0$. The second is the connecting map associated to the sequence $0 \to A[2] \to A^\chi[4] \xrightarrow{2} A[2] \to 0$. We denote the second by $\delta_\chi : A(k_v)[2] \to H^1(k, A[2])$.

Lemma 4.5. *For $x \in A(k_v)[2]$ we have $\delta_\chi(x) = \delta(x) + \chi \cup x$.*

Proof. Let $y \in A[4]$ be such that $2y = x$. Then for all $\sigma \in G_k$, we have

$$\delta_\chi(x)(\sigma) = \phi^{-1}(\sigma\phi(y) - \phi(y)) = \chi(\sigma)\sigma y - y = \begin{cases} \delta(x)(\sigma) & \chi(\sigma) = 1, \\ \delta(x)(\sigma) + x & \chi(\sigma) = -1, \end{cases}$$

from which the result follows. □

Lemma 4.6. *Let $v \nmid 2$ be a nonarchimedean place of good reduction for A. Let $\chi : G_k \to \mu_2$ be a quadratic character.*

(i) *If χ is unramified at v then*

$$\delta(A(k_v)) = H^1_{\mathrm{ur}}(k_v, A[2]) = \delta(A^\chi(k_v)).$$

(ii) *If χ is ramified at v then we have $\delta(A^\chi(k_v)) = \delta_\chi(A(k_v)[2])$. Moreover, we have*

$$\delta(A(k_v)) \cap \delta^\chi(A(k_v)) = 0.$$

Proof. (i). Since χ is unramified at v, both A and A^χ have good reduction at v and the result follows from Lemma 4.4 (ii).

(ii). Since A has good reduction at v, $A[4]$ is unramified. On the other hand, since χ is ramified at v, there is σ in the inertia group I_v at v such that $\chi(\sigma) = -1$. Then σ acts a multiplication by -1 on $A^\chi[4]$. From this, we deduce that $H^0(k_v, A^\chi[4]) = A(k_v)[2]$. Thus, $\delta_\chi : A(k_v^{\mathrm{ur}})[2] \to H^1(k^{\mathrm{ur}}, A[2])$ is injective. In particular, $\delta_\chi : A(k_v)[2] \to H^1(k_v, A[2])$ is injective. Lemma 4.4 (ii) gives $\#\delta(A^\chi(k_v)) = \#A(k_v)[2]$, so we deduce that $\delta(A^\chi(k_v)) = \delta_\chi(A(k_v)[2])$. Finally, we have $\delta(A(k_v)) = H^1_{\mathrm{ur}}(k_v, A[2])$ so we must show that $H^1_{\mathrm{ur}}(k_v, A[2]) \cap \delta_\chi(A(k_v)[2]) = 0$. This follows from the injectivity of $\delta_\chi : A(k_v^{\mathrm{ur}})[2] \to H^1(k^{\mathrm{ur}}, A[2])$ noted previously. □

Let S be a finite set of places of k and let $\chi : G_k \to \mu_2$ be a quadratic character. Write

$$V_S = \mathrm{im}\left(\mathrm{Sel}_2(A) \overset{\oplus_{v\in S}\mathrm{res}_v}{\longrightarrow} \bigoplus_{v\in S} \frac{\delta(A(k_v))}{\delta(A(k_v)) \cap \delta(A^\chi(k_v))}\right),$$

and define V_S^χ similarly by interchancing A and A^χ in the definition. The following result is due to Harpaz [Har19a, Lemma 3.27], building on work of Mazur–Rubin [MR10, Section 3].

Proposition 4.7. *Suppose that $\delta(A(k_v)) = \delta(A^\chi(k_v))$ for all $v \notin S$. Then*

$$\dim_{\mathbb{F}_2} \mathrm{Sel}_2(A^\chi) = \dim_{\mathbb{F}_2} \mathrm{Sel}_2(A) + \dim_{\mathbb{F}_2} V_S^\chi - \dim_{\mathbb{F}_2} V_S. \tag{4.8}$$

Moreover, we have

$$\dim_{\mathbb{F}_2} V_S + \dim_{\mathbb{F}_2} V_S^\chi \leq \sum_{v\in S} \dim_{\mathbb{F}_2} \frac{\delta(A(k_v))}{\delta(A(k_v)) \cap \delta(A^\chi(k_v))}.$$

Proof. To ease notation, for each place v if k, write $\mathcal{L}_v = \delta(A(k_v))$ and $\mathcal{F}_v = \delta(A^\chi(k_v))$. Define a Selmer structure $\mathcal{G}$ for $A[2]$ by setting $\mathcal{G}_v = \mathcal{L}_v \cap \mathcal{F}_v$ for each place v. We have evident exact sequences

$$0 \to \mathrm{Sel}_{\mathcal{L}}(A[2]) \to \mathrm{Sel}_2(A) \overset{\oplus_{v\in S}\mathrm{res}_v}{\to} V_S \to 0$$

and

$$0 \to \mathrm{Sel}_{\mathcal{L}}(A[2]) \to \mathrm{Sel}_2(A^\chi) \overset{\oplus_{v\in S}\mathrm{res}_v}{\to} V_S^\chi \to 0.$$

Taking $\mathbb{F}_2$-dimensions gives Equation (4.8).

For each place v, since both $\mathcal{F}_v$ and $\mathcal{L}_v$ are their own orthogonal complements with respect to the local Tate pairing, we have a non-degenerate pairing

$$\mathcal{L}_v/\mathcal{G}_v \times F_v/\mathcal{G}_v \to \tfrac{1}{2}\mathbb{Z}/\mathbb{Z}$$

given by restricting the local Tate pairing. Summing over $v \in S$ gives a non-degenerate pairing

$$\oplus_{v\in S}\mathcal{L}_v/\mathcal{G}_v \times \oplus_{v\in S}\mathcal{F}_v/\mathcal{G}_v \to \tfrac{1}{2}\mathbb{Z}/\mathbb{Z}. \tag{4.9}$$

For each $x \in \mathrm{Sel}_2(A)$ and $y \in \mathrm{Sel}_2(A^\chi)$, reciprocity for the Brauer group of k gives

$$\sum_{v \text{ place of } k} \mathrm{inv}_v(x \cup y) = 0.$$

Further, since the Selmer conditions for A and A^χ agree for all places $v \notin S$, we have $\mathrm{inv}_v(x \cup y) = 0$ for all $v \notin S$. Thus, we deduce that V_S and V_S^χ are orthogonal with respect to the pairing (4.9). In particular, we have

$$\dim_{\mathbb{F}_2} V_S + \dim_{\mathbb{F}_2} V_S^\chi \leq \dim_{\mathbb{F}_2} \oplus_{v\in S}\mathcal{L}_v/\mathcal{G}_v,$$

from which the result follows. □

Remark 4.10. One can strengthen Proposition 4.7 by showing that we moreover have

$$\dim_{\mathbb{F}_2} V_S + \dim_{\mathbb{F}_2} V_S^\chi \equiv \sum_{v \in S} \dim_{\mathbb{F}_2} \frac{\delta(A(k_v))}{\delta(A(k_v)) \cap \delta(A^\chi(k_v))} \pmod 2.$$

This is more subtle to prove and uses the existence of certain quadratic refinements of the local Tate pairing arising from Mumford theta groups; see [Har19a, Lemma 3.27] for details. However, it can be very useful in practice, in particular if one wants to twist to increase the 2-Selmer group – it sometimes lets one deduce that $\dim_{\mathbb{F}_2} V_S^\chi > 0$ just by understanding V_S.

Corollary 4.11. *Let Σ be a finite set of places of k containing all archimedean places, all places dividing 2 and all places of bad reduction for A. Suppose that $\chi : G_k \to \mu_2$ is a quadratic character that is trivially locally at all places in Σ. Suppose that χ is ramified at a single place $v \notin \Sigma$ and that the restriction map $\mathrm{res}_v : \mathrm{Sel}_2(A) \to H^1_{\mathrm{ur}}(k_v, A[2])$ is surjective. Then*

$$\dim_{\mathbb{F}_2} \mathrm{Sel}_2(A^\chi) = \dim_{\mathbb{F}_2} \mathrm{Sel}_2(A) - \dim_{\mathbb{F}_2} A(k_v)[2].$$

Proof. By Lemma 4.6(i) and our assumptions on χ, we may take $S = \{v\}$ in Proposition 4.7. Using Lemma 4.4(ii) and the proposition, we have $\dim V_S + \dim V_S^\chi \leq \dim A(k_v)[2]$. Moreover, Lemma 4.4 and the assumption that the restriction map is surjective shows that $V_S = H^1_{\mathrm{ur}}(k_v, A[2])$. Now $\dim H^1_{\mathrm{ur}}(k_v, A[2]) = \dim A(k_v)[2]$ so we deduce from the inequality that $V_S^\chi = 0$, from which the result follows. □

4.4. **Parity of Selmer ranks.** A predicted consequence of the Birch and Swinnerton-Dyer conjecture is the parity conjecture:

$$(-1)^{\mathrm{rk} A} = w(A).$$

Here the right hand side is the *global root number* of A. By definition, $w(A) \in \{\pm 1\}$ is a product of *local root numbers*:

$$w(A) = \prod_{v \text{ place of } k} w(A/k_v).$$

One expects that $w(A)$ is the sign in the (conjectural) functional equation for the L-function of A, hence controls the parity of the order of vanishing of the L-function at $s = 1$, which is why the parity conjecture is predicted by the Birch and Swinnerton-Dyer conjecture. This means that the parity of $\mathrm{rk} A$ should be determined by quite simple local invariants of A, namely the local root numbers $w(A/k_v) \in \{\pm 1\}$. We remark that if A has good reduction at v then $w(A/k_v) = 1$.

Now let χ be a quadratic character with corresponding quadratic extension F/k. We have

$$\mathrm{rk}(A/F) = \mathrm{rk}(A/k) + \mathrm{rk}(A^\chi/k).$$

In particular, the parity conjecture predicts that

$$(-1)^{\mathrm{rk}(A^\chi/K)} = (-1)^{\mathrm{rk} A} \prod_{w \text{ place of } F} w(A/F_w).$$

Since $w(A/F_w) = 1$ whenever w lies over a place of good reduction for A, we see that the parity of $\mathrm{rk}(A^\chi/k)$ depends on χ only through its restriction to k_v for the finitely many places v of k at which A has bad reduction. Assuming finiteness of Shafarevich–Tate groups, a version of this can be proven unconditionally.

Theorem 4.12 ([Mor19], Theorem 10.12)**.** *Let Σ be a finite set of places of k consisting of all archimedean places, all places dividing 2 and all places at which A has bad reduction. For each $v \in \Sigma$, let $\chi_v : G_{k_v} \to \{\pm 1\}$ be a local quadratic character. Assume that the Shafarevich–Tate group of each quadratic twist of A is finite. Then there are local invariants $\Omega_v(\chi_v) \in \{\pm 1\}$ such that, for all global quadratic characters $\chi : G_k \to \{\pm 1\}$, we have*

$$(-1)^{\mathrm{rk} A^\chi} = (-1)^{\mathrm{rk} A} \prod_{v \in \Sigma} \Omega_v(\mathrm{res}_v(\chi)).$$

While it is not known that the local invariants κ_v agree with the corresponding local root numbers in general, it is known in some cases. It what follows it will be useful to know that A has a quadratic twist of odd (hence positive) rank, at least assuming finiteness of Shafarevich–Tate groups. Part (i) of the following result, which is predicted by root numbers, serves to accomplish this in many instances.

Proposition 4.13. *Suppose $v \nmid 2\infty$ is a place of k such that A has semistable reduction at v, and such that the group of geometric connected components of the special fibre of the Néron model of A over k_v has odd order. Let χ be the unique non-trivial unramified quadratic character of G_{k_v}.*

(i) *We have*
$$\Omega_v(\chi) = (-1)^{\mathfrak{f}(A/k_v)},$$
where $\mathfrak{f}(A/k_v)$ denotes the conductor exponent of A at v.

(ii) *We have $\delta(A(k_v)) = H^1_{\mathrm{ur}}(k_v, A[2]) = \delta(A^\chi(k_v))$.*

Proof. (i) This is [Mor25], Proposition 6.3.

(ii). Since Néron models commute with unramified base change, it suffices to prove this for $\delta(A(k_v))$. Since $v \nmid 2\infty$, we have
$$\dim H^1_{\mathrm{ur}}(k_v, A[2]) = \dim A(k_v)[2] = \dim \delta(A(k_v)),$$
hence it suffices to show that $H^1_{\mathrm{ur}}(k_v, A[2])$ is contained in $\delta(A(k_v))$. Equivalently, we want to show that the image of $H^1_{\mathrm{ur}}(k_v, A[2])$ in $H^1(k_v, A)$ is trivial. By [Mil06, Proposition I.3.8] we have
$$H^1_{\mathrm{ur}}(k_v, A) \cong H^1_{\mathrm{ur}}(k_v, \Phi),$$
where Φ is the group of geometric connected components of the special fibre of the Néron model. Since Φ has odd order, it follows that $H^1_{\mathrm{ur}}(k_v, A)[2]$ is trivial, hence so is the image of $H^1_{\mathrm{ur}}(k_v, A[2])$ in $H^1(k_v, A)$. □

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